Universal morphisms

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https://soham-saha.github.io

https://soham-saha.codeberg.page

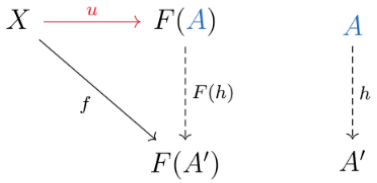
Let us have a look at Wikipedia:

Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between categories $\mathcal{C}$ and $\mathcal{D}$. In what follows, let X be an object of $\mathcal{D}$, A and A' be objects of $\mathcal{C}$, and $h : A \to A'$ be a morphism in $\mathcal{C}$.

Then, the functor F maps A, A' and h in $\mathcal{C}$ to $F(A)$, $F(A')$ and $F(h)$ in $\mathcal{D}$.

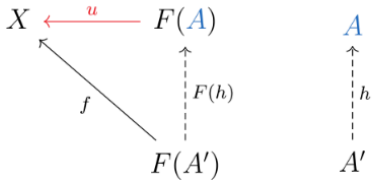
A **universal morphism from X to F** is a unique pair $(A, u : X \to F(A))$ which has the following property, commonly referred to as a **universal property**:

For any morphism of the form $f : X \to F(A')$ in $\mathcal{D}$, there exists a *unique* morphism $h : A \to A'$ in $\mathcal{C}$ such that the following diagram commutes:



We can dualize this categorical concept. A **universal morphism from F to X** is a unique pair $(A, u : F(A) \to X)$ that satisfies the following universal property:

For any morphism of the form $f : F(A') \to X$ in $\mathcal{D}$, there exists a *unique* morphism $h : A' \to A$ in $\mathcal{C}$ such that the following diagram commutes:



In the first case (universal morphism from X to F), the pair $(A, u : X \to F(A))$ is universal in the sense that any pair $(A', u' : X \to F(A'))$ factors uniquely through $h : A \to A'$.

We can also reconcile this idea with comma categories, by saying that (A, u) is initial in the comma category $!_X \downarrow F$.

Dually speaking, universal morphisms from F to X correspond to terminal objects in the comma category $F \downarrow !_X$.